Diffusion Limited Aggregation

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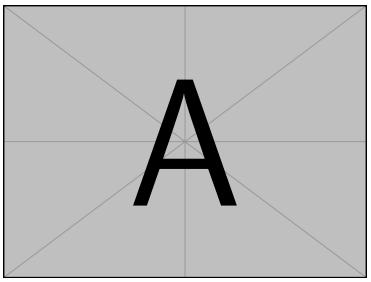

A dissertation submitted for the degree of
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Abstract

This dissertation investigates the universality class of Diffusion Limited Aggregation (DLA) through the lens of conformal mapping and harmonic measure theory. We compare on-lattice, off-lattice, and hybrid "snapping" algorithms to determine if microscopic lattice anisotropy survives in the macroscopic scaling limit. Particular attention is paid to the dimension of the harmonic measure and the dynamics of the active growth zone. [Placeholder: Finish this summary with your specific results regarding the Hybrid model's classification, totaling approx. 300 words.]

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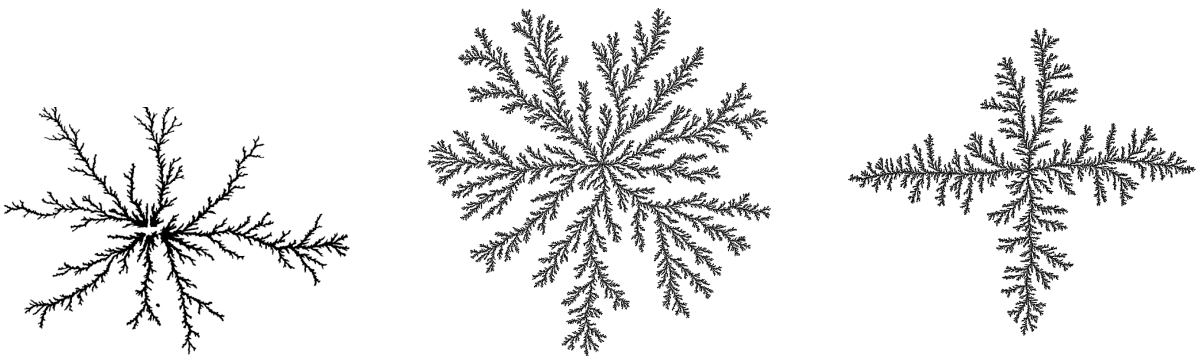
Chapter 1

Introduction

1.1 Overview

Diffusion Limited Aggregation (DLA) is a stochastic model for irreversible colloidal aggregation. In this model a seed particle is placed at the origin, and then subsequent particles are released from a boundary far from the cluster one at a time. The particle then diffuses, either as a random walk on a lattice or Brownian motion (the continuous analogue to a random walk). Upon collision, the walker sticks and the next one starts diffusing.

The model was first proposed in 1981 by Witten and Sander to model the colloidal aggregation of metal precipitates [1], but has since been found to have applications in a diverse range of physical systems, including electrodeposition [2], bacteria growth [3] and viscous fingering in Hele-Shaw cells [4] among others.



(a) Zinc Electrodeposition [2] **(b)** Off-Lattice DLA **(c)** On-Lattice DLA

Figure 1.1: Comparison of physical aggregation (a) with the two primary modelling approaches: continuum Off-Lattice DLA (b) and discrete On-Lattice DLA (c). Both are the results of simulation with 10^7 particles. Note the distinct grid-induced anisotropy visible in the on-lattice case.

1.2 Theoretical Context

While DLA is defined by a stochastic algorithm (random walkers), its theoretical significance lies in its connection to potential theory, the study of harmonic functions. As will be detailed in the following chapters, the probability density $\phi(u)$ of a diffusing particle at position u satisfies the Laplace equation $\nabla^2\phi = 0$. Intuitively, this is because both random walks and harmonic

functions are defined by the property that the value at any point is the average of its neighbours. Each additional walker to aggregate onto the cluster distorts this probability field, creating a Laplacian instability that amplifies irregularities into the observed complicated structure [5].

1.3 Fractal Dimension

Visually, DLA clusters exhibit self-similarity, where small branches look statistically identical to large branches. Mathematically this can be understood via fractal dimension D. This will be formally defined later, but can be thought of as the relationship between the mass of a cluster (number of particles N) and its linear size R, where:

$$N \propto R^D. \tag{1.1}$$

For instance, doubling the size of a line would double its mass, so it has dimension 1, while doubling the size of a solid disc would quadruple its mass, so this has dimension 2. Simulations of off-lattice DLA have consistently yielded a fractal dimension $D \approx 1.71$, but this has not been proven [6].

In fact, the only mathematical theorem on 2D DLA is due to Kesten, who proved a bound on the growth rate of on-lattice DLA in 1987, with a key consequence being that the fractal dimension of a DLA cluster is bounded below by 1.5 [7].

1.4 Anisotropy and Thesis Goals

It is likely that the dimension of on-lattice DLA is actually lower than off-lattice due to the effect of the grid on the cluster shape, a phenomenon known as anisotropy. Grebenkov and Beliaev discussed how anisotropy is likely to reduce the dimension of large enough clusters [8], but understanding why and to what extent is still not fully understood.

To answer these questions, simulations of DLA must be done both on and off-lattice. Additionally, to understand the effect of the lattice we create a 'hybrid' algorithm with continuous diffusion but a discrete aggregation method. Other hybrid models of DLA have previously been studied [9], including a similar semi-lattice approach [10], but to our knowledge have not been utilised to decouple the effects of transport from aggregation in on-lattice DLA.

The purpose of this dissertation is to compare these three different models of DLA, in order to decouple the diffusion process of DLA from the geometry of the aggregation. By analysing various features of the resulting clusters, we hope to better understand the relationship between the continuous and discrete variations of the probability distribution governing DLA growth.

Chapter 2

Potential Theory

2.1 Preliminaries

2.1.1 Martingales and the Optional Stopping Theorem

In this section, we review the probabilistic tools required to link random walks to potential theory. We assume standard measure-theoretic knowledge (equivalent to the Part A Integration course).

Definition 2.1 (Probability Space)**.** *A probability space* $(\Omega, \mathcal{F}, \mathbb{P})$ *is a triple where* Ω *is a sample set,* $\mathcal{F}$ *is a* σ*-algebra on* Ω *representing the set of events, and* $\mathbb{P}\colon \mathcal{F} \to [0,1]$ *is a countably additive set function such that* $\mathbb{P}(\Omega) = 1$*.*

Definition 2.2 (Filtration)**.** *A filtration on a probability space* $(\Omega, \mathcal{F}, \mathbb{P})$ *is a sequence of sub-*σ*-algebras* $(\mathcal{F}_n)_{n\geq 0}$ *such that:*

$$\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \mathcal{F}_2 \subseteq \cdots \subseteq \mathcal{F}.$$

The tuple $(\Omega, \mathcal{F}, (\mathcal{F}_n)_{n\geq 0}, \mathbb{P})$ *is called a filtered probability space.*

Intuition: The filtration $(\mathcal{F}_n)$ represents the accumulation of information over time. $\mathcal{F}_n$ contains all events whose occurrence (or non-occurrence) is known by time n.

Definition 2.3 (Stopping Time)**.** *A random variable* $\tau\colon \Omega \to \{0, 1, \ldots, \infty\}$ *is called a stopping time with respect to the filtration* $(\mathcal{F}_n)_{n\geq 0}$ *if for every* $n \geq 0$*:*

$$\{\omega \in \Omega : \tau(\omega) \leq n\} \in \mathcal{F}_n.$$

Intuition: A stopping time represents a strategy where the decision to stop at time n depends only on the history of the process up to time n. For example, "stop when the walker hits the cluster" is a valid stopping time but "stop at the step before the walker hits the cluster" is not.

Definition 2.4 (Martingale)**.** *A discrete-time stochastic process* $(X_n)_{n\geq 0}$ *is a martingale with respect to a filtration* $(\mathcal{F}_n)_{n\geq 0}$ *if:*

1. X_n *is adapted to the filtration (i.e.,* X_n *is* $\mathcal{F}_n$*-measurable for all* n*),*
2. $\mathbb{E}[|X_n|] < \infty$ *for all* n *(integrability),*
3. $\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] = X_n$ *almost surely (conditional expectation).*

Intuition: A martingale models a “fair game" with no drift. The condition $\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] = X_n$ implies that given the entire history of the process, the best prediction for the next value is the current value.

Theorem 2.1 (Doob's Optional Stopping Theorem)**.** *Let X be a martingale on a filtered probability space. If τ is a stopping time such that $\mathbb{P}(\tau < \infty) = 1$, and X is uniformly bounded (i.e., there exists K such that $|X_n(\omega)| \leq K$ for all n, ω), then:*

$$\mathbb{E}[X_\tau] = \mathbb{E}[X_0]. \tag{2.1}$$

Proof. See Williams [11, Section 10.9]. □

Intuition: The Optional Stopping Theorem states that for a fair game, no stopping strategy τ can change the expected payout. This will allow us to relate the value of a harmonic function at a starting point x (start of the walk) to its expected value at the boundary ∂D (end of the walk).

2.1.2 Measure Theory Concepts

In the study of potential theory, we will want to understand the relationship between the geometric length of a boundary (Lebesgue measure) and the probability that a random walker hits that boundary (harmonic measure). To rigorously define a probability density function linking these two concepts we require the notion of the Radon-Nikodym derivative.
Remark: throughout this work, we assume our measurable spaces are equipped with the Borel σ-algebra.

Definition 2.5 (Absolute Continuity)**.** *Let $(\Omega, \mathcal{F})$ be a measurable space equipped with two measures μ and ν. We say that ν is absolutely continuous with respect to μ, denoted $\nu \ll \mu$, if for every $A \in \mathcal{F}$:*

$$\mu(A) = 0 \implies \nu(A) = 0. \tag{2.2}$$

Intuition: If $\nu \ll \mu$, then ν assigns zero mass to any set where μ assigns zero mass. This ensures that we can't have a "probability" of being in a region with zero "volume."

Theorem 2.2 (The Radon-Nikodym Theorem)**.** *Let $(\Omega, \mathcal{F})$ be a measurable space, and let μ and ν be two finite measures such that $\nu \ll \mu$. Then, there exists a non-negative measurable function $f \colon \Omega \to [0, \infty)$, unique μ-almost everywhere, such that for any $A \in \mathcal{F}$:*

$$\nu(A) = \int_A f \, d\mu. \tag{2.3}$$

The function f is called the Radon-Nikodym derivative of ν with respect to μ and is denoted by:

$$f = \frac{d\nu}{d\mu}. \tag{2.4}$$

Proof. See Williams [11, Section 5.14]. □

2.1.3 Conformal Maps

Here we state the definition of a conformal map, and the Riemann mapping theorem. Conformal maps will be very important for two reasons: theoretically, the Riemann Mapping Theorem guarantees that the Dirichlet problem is well-posed even on the fractal boundary of a DLA cluster; algorithmically, these maps allow us to exploit the 'conformal invariance' of Brownian motion (discussed later)

Definition 2.6 (Conformal Map)**.** *Let $U, V \subseteq \mathbb{C}$ be open sets. A function $f\colon U \to V$ is called* ***conformal*** *at a point $z_0 \in U$ if it preserves angles and orientation locally.*

Analytically, a holomorphic function f is conformal at every point where its derivative is non-zero, i.e., $f'(z) \neq 0$. If f is a holomorphic bijection between U and V (a biholomorphism), it is essentially a global conformal map, and we say U and V are conformally equivalent.

Theorem 2.3 (Riemann Mapping Theorem)**.** *Let $U \subsetneq \mathbb{C}$ be a non-empty, simply connected open subset of the complex plane (i.e., U is the whole plane minus at least one point, with no holes). Then, there exists a unique bijective conformal map f from U onto the open unit disk $\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}$ such that:*

$$f(z_0) = 0 \quad \textit{and} \quad f'(z_0) > 0, \tag{2.5}$$

for any prescribed point $z_0 \in U$.

Proof. See Ahlfors [12, Chapter 6] or Priestley [13, Chapter 9]. □

2.2 Discrete Potential Theory[1]

We now introduce potential theory in a discrete setting, focusing on its applications to DLA. We start by considering the definitions of a random walk and a discrete harmonic function.

2.2.1 Simple Random Walks and the Generator

Definition 2.7 (Random Walk)**.** *A random walk on $\mathbb{Z}^2$ is a stochastic process $(S_n)_{n\geq 0}$ defined by $S_n = S_0 + \sum_{i=1}^{n} X_i$, where $X_1, X_2, \ldots$ are independent and identically distributed random variables taking values in $\mathbb{Z}^2$.*

The common distribution of the steps is denoted by $p(x) = \mathbb{P}(X_i = x)$ and is called the ***increment distribution****.*

For the purposes of DLA, we are primarily interested in the symmetric case:

Definition 2.8 (Simple Symmetric Random Walk)**.** *A random walk is called a Simple Symmetric Random Walk (SSRW) if the increment distribution is uniform on the nearest neighbors:*

$$p(x) = \begin{cases} \frac{1}{4} & \textit{if } |x| = 1, \\ 0 & \textit{otherwise.} \end{cases} \tag{2.6}$$

[1] The theoretical framework in this section is largely adapted from Lawler and Limic [5], applied here to the specific case of 2D DLA.

We can now define the operator that governs the evolution of the expectation of functions under this flow.

Definition 2.9 (Discrete Generator)**.** *Let $p(\cdot)$ be the increment distribution of a random walk. The generator $\mathcal{L}$ is a linear operator acting on functions $f : \mathbb{Z}^2 \to \mathbb{R}$ defined by:*

$$\mathcal{L}f(x) = \sum_{y \in \mathbb{Z}^2} p(y - x)[f(y) - f(x)]. \tag{2.7}$$

Probabilistically, this represents the expected change in the function f in a single step of the walk starting at x:

$$\mathcal{L}f(x) = \mathbb{E}[f(S_1) - f(S_0) \mid S_0 = x]. \tag{2.8}$$

Remark: For Simple Symmetric Random Walk, the generator is proportional to the unnormalized discrete Laplacian, $\mathcal{L} = \frac{1}{4}\Delta$, where $\Delta f(x) = \sum_{y \sim x}(f(y) - f(x))$.

Definition 2.10 (Discrete Harmonic Functions)**.** *A function $f : A \cup \partial A \to \mathbb{R}$ is said to be harmonic in a subset $A \subset \mathbb{Z}^2$ if:*

$$\mathcal{L}f(x) = 0 \quad \textit{for all } x \in A. \tag{2.9}$$

Proposition 2.1 (Harmonic Functions are Martingales)**.** *Let S_n be a random walk and f be a function harmonic on a set A. Let $\tau_A = \inf\{n \geq 0 : S_n \notin A\}$ be the first exit time from A. Then the stopped process $M_n = f(S_{n \wedge \tau_A})$ is a martingale (with respect to the natural filtration of S_n).*

Proof. See Lawler and Limic [5, Proposition 6.1.1]. □

2.2.2 The Discrete Dirichlet Problem

Definition 2.11 (Boundary)**.** *Before proceeding, we distinguish between the occupied cluster and the available growth sites. Let $K \subset \mathbb{Z}^2$ denote the set of occupied sites (the cluster). We define the boundary ∂K as the outer boundary consisting of all empty sites adjacent to the cluster:*

$$\partial K = \{y \in \mathbb{Z}^2 \setminus K : \exists x \in K \textit{ such that } |y - x| = 1\}. \tag{2.10}$$

Subsequently we refer to the outer boundary as the ***boundary*** *of the cluster ∂K.*

We now attempt to model DLA as a discrete Dirichlet problem, where the unknown function $u_R(x)$ is defined as the probability of a random walker starting at x escaping to some outer boundary ∂R of radius R before hitting the cluster boundary ∂K.

$$u_R(x) = \mathbb{P}^x(\text{hit outer boundary } \partial R \text{ before } \partial K). \tag{2.11}$$

As the diffusion process is a simple random walk, the probability of escaping from x is simply the average of the excape probabilities of its neighbours:

$$u_R(x) = \frac{1}{4} \sum_{z \sim x} u_R(z). \tag{2.12}$$

By definition this makes $u(\cdot)$ harmonic, hence satisfies the discrete Laplace equation $\mathcal{L}u(x) = 0$.

Thus, each stage of DLA is equivalent to solving the Dirichlet problem:

$$\begin{cases} \Delta f(x) = 0 & \text{if } x \notin K \text{ (Harmonic in free space)}, \\ f(x) = 0 & \text{if } x \in \partial K \text{ (Hit the cluster)}, \\ f(x) = 1 & \text{if } x \in \partial R \text{ (Escaped to infinity)}. \end{cases} \tag{2.13}$$

Remark 2.1. *We could have solved for the probability of hitting the cluster before escaping, reversing the boundary conditions. However, we do it this way to preserve consistency with the later sections and the electrostatic interpretation (see Appendix).*

We now look at the general discrete Dirichlet problem in 2D:

Theorem 2.4 (Solution to the 2D Discrete Dirichlet Problem)**.** *Let $A \subset \mathbb{Z}^2$ be a subset (potentially infinite) and let $\tau = \inf\{n \geq 0 : S_n \notin A\}$ be the first exit time. Assume that:*

1. *The random walk stops almost surely: $\mathbb{P}_x(\tau < \infty) = 1$ for all $x \in A$.*
2. *The boundary function $g : \partial A \to \mathbb{R}$ is bounded.*

Then, the function $f : A \cup \partial A \to \mathbb{R}$ defined by the expectation:

$$f(x) = \mathbb{E}_x[g(S_\tau)] \tag{2.14}$$

is the unique bounded solution to the Dirichlet system:

$$\mathcal{L}f(x) = 0, \qquad \text{for all } x \in A, \tag{2.15a}$$

$$f(x) = g(x), \qquad \text{for all } x \in \partial A. \tag{2.15b}$$

Proof. **Existence:** We first check that (2.14) satisfies the system (2.15). If $x \in \partial A$, then $\tau = 0$, so $S_\tau = x$ and $f(x) = \mathbb{E}_x[g(x)] = g(x)$. Thus the boundary condition (2.15b) holds. If $x \in A$, we use the Markov property at the first step:

$$f(x) = \mathbb{E}_x[g(S_\tau)] = \sum_{y \sim x} p(y - x)\mathbb{E}_y[g(S_\tau)] = \sum_{y \sim x} p(y - x)f(y). \tag{2.16}$$

Thus $\mathcal{L}f(x) = 0$, satisfying (2.15a).

Uniqueness: Suppose h is another bounded solution to the system (2.15). Since h satisfies (2.15a), it is harmonic, and by proposition 2.1, the process $M_n = h(S_{n\wedge\tau})$ constitutes a martingale. Since h is bounded, M_n is a bounded martingale.
Given that $\tau < \infty$ almost surely, we can apply the Optional Stopping Theorem (theorem 2.1):

$$h(x) = \mathbb{E}_x[M_0] = \mathbb{E}_x[M_\tau] = \mathbb{E}_x[h(S_\tau)]. \tag{2.17}$$

From condition (2.15b), we know $h(S_\tau) = g(S_\tau)$ on the boundary. Therefore:

$$h(x) = \mathbb{E}_x[g(S_\tau)], \tag{2.18}$$

which is exactly $f(x)$. Thus $h \equiv f$, proving uniqueness. □

2.2.3 Recurrence and the Potential Kernel

To apply this solution to DLA, we require taking the limit $R \to \infty$, which is equivalent to solving the Dirichlet problem with a boundary condition at infinity.

This is done formally by Lawler and Limic [5, Proposition 6.2.2], who prove that a solution to the Dirichlet problem in d dimensions is uniquely determined by values on the finite boundary ∂A and the value at infinity, taking the form:

$$f(x) = \mathbb{E}_x[g(S_\tau)\mathbf{1}_{\tau<\infty}] + g(\infty)\mathbb{P}_x(\tau = \infty). \tag{2.19}$$

However, due to the recurrence of a random walk in 2D space, the escape probability $\mathbb{P}_x(\tau = \infty)$ is zero for any random walk in $\mathbb{Z}^2$. Consequently, the second term vanishes. The solution is now entirely determined by the inner boundary ∂K. For our boundary conditions given in eq. (2.15), the only solution is the trivial potential $f \equiv 0$.

To address this we now turn to the concept of Green's functions. The Green's function $G(x, y)$ represents the potential field at x induced by a point source at y. Put another way, it is the solution to the Poisson equation:

$$\Delta G(x, y) = -\delta_y(x). \tag{2.20}$$

In the probabilistic setting, the Green's function has a precise interpretation: it is the expected number of times a random walk starting at x visits the site y:

$$G(x, y) = \mathbb{E}_x\left[\sum_{n=0}^{\infty} \mathbf{1}_{\{S_n=y\}}\right] = \sum_{n=0}^{\infty} p_n(y - x). \tag{2.21}$$

For transient random walks (such as in dimensions $d \geq 3$), this sum converges, and $G(x, y)$ decays to zero as $|x - y| \to \infty$.

However, in two dimensions the random walk is recurrent, meaning a walker returns to the origin infinitely often, causing the sum in the definition of $G(x, y)$ to diverge:

$$G(x, y) = \infty \quad \text{for all } x, y. \tag{2.22}$$

Instead we regularise the divergent Green's function by working with potential differences, motivating the definition of the potential kernel.

Definition 2.12 (Potential Kernel)**.** *The potential kernel $a(x)$ for simple random walk in $\mathbb{Z}^2$ is defined as:*

$$a(x) = \sum_{n=0}^{\infty}[p_n(0) - p_n(x)], \tag{2.23}$$

where $p_n(x) = \mathbb{P}(S_n = x)$.

It is not immediately clear that this will be finite. However, using estimates from the Local Central Limit Theorem [5, Section 4.4.1], it can be shown that for any fixed x, the difference decays as:

$$|p_n(0) - p_n(x)| \leq C|x|n^{-3/2}. \tag{2.24}$$

Since the series $\sum_{n=1}^{\infty} n^{-3/2}$ converges absolutely, the potential kernel $a(x)$ is well-defined and

finite for all x.

By transitioning from the Dirichlet problem to the potential kernel, we stop solving directly for the probability and instead solve for a renormalised potential $a(x)$. Intuitively, $a(x)$ quantifies the relative accessibility of a given lattice site from infinity. Importantly, the growth probability is recovered through the discrete differences of this potential: The proportion of walkers hitting a boundary point depends on the 'steepness' of the potential at the boundary.
The potential kernel therefore acts as the fundamental solution to the Laplacian on the lattice. It satisfies the Poisson equation:

$$\mathcal{L}a(x) = \delta_0(x) = \begin{cases} 1 & x = 0, \\ 0 & x \neq 0. \end{cases} \tag{2.25}$$

It replaces the 'boundary at infinity' with an asymptotic growth condition.

Theorem 2.5 (Asymptotics of the Potential Kernel)**.** *As $|x| \to \infty$, the potential kernel satisfies:*

$$a(x) = \frac{2}{\pi} \ln |x| + k + O(|x|^{-2}), \tag{2.26}$$

where k is a constant involving Euler's constant γ.

Proof. See Lawler and Limic [5, Theorem 4.4.4]. □

Remark 2.2. *The logarithmic growth may seem arbitrary, but can in fact be justified with some geometric intuition: Consider a large circle of radius R centered on the cluster (large so that it contains the entire cluster). For a steady flow of random walkers coming from infinity to reach the cluster, the total flux (rate of flow) across this circle must be independent of R. The total flux is the product of the circumference ($2\pi R$) and the radial gradient of the potential ($\nabla f(R)$). This implies that the gradient must decay as $\nabla f(R) \propto 1/R$. Integrating this gradient yields:*

$$f(R) \propto \int \frac{1}{R}\, dR = \ln R. \tag{2.27}$$

Non-logarithmic growth would imply that the flux across a circle varies depending on its radius, violating physical conservation laws.

2.2.4 Discrete Harmonic Measure

We now formally define the discrete harmonic measure:

Definition 2.13 (Harmonic Measure from Infinity)**.** *Let $K \subset \mathbb{Z}^2$ be a finite cluster. The harmonic measure $H_{\partial K}(y)$ is the probability that a simple random walk starting from infinity first hits the cluster K at the point $y \in \partial K$. Formally, it is defined as the limit:*

$$H_{\partial K}(y) = \lim_{|x| \to \infty} \mathbb{P}_x(S_{\tau_K} = y), \tag{2.28}$$

where $\tau_K = \inf\{n \geq 0 : S_n \in K\}$ is the first hitting time.

Lawler and Limic [5, Proposition 6.6.1] prove that this limit exists and defines a unique probability measure on the boundary ∂K. While the definition may seem abstract, we can in fact calculate these probabilities directly from the potential field derived in the previous section.

We begin by formally defining the concept of discrete flux which was invoked in remark 2.2. On the lattice, the concept of a normal derivative $\nabla u \cdot \mathbf{n}$ is replaced by a difference operator across the boundary.

Definition 2.14 (Discrete Flux)**.** *For a function $u : \mathbb{Z}^2 \to \mathbb{R}$ that vanishes on K, the discrete flux at a boundary point $y \in \partial K$ is the sum of the potential differences from its neighbors in the complement:*

$$\Phi(y) = \sum_{z \sim y, z \notin K} (u(z) - u(y)). \tag{2.29}$$

Since $u(y) = 0$ on the boundary, this simplifies to the sum of the external neighbors: $\Phi(y) = \sum_{z \sim y, z \notin K} u(z)$.

The fundamental result that we require from discrete potential theory is the equivalence of flux and the harmonic measure.

Proposition 2.2 (Harmonic Measure as Flux)**.** *The harmonic measure at $y \in \partial K$ is proportional to the discrete flux at y:*

$$H_{\partial K}(y) = \frac{\Phi(y)}{\sum_{w \in \partial K} \Phi(w)}. \tag{2.30}$$

Proof. For a formal rigorous proof connecting hitting probabilities to the Laplacian of the Green's function, see Lawler and Limic [5, Proposition 1.6.3]. □

To understand this link intuitively, we interpret the potential $u(x)$ as the density of a stream of random walkers originating from infinity and disappearing upon hitting the cluster K. In the exterior free space, the condition $\mathcal{L}u(x) = 0$ implies a sort of conservation law: the expected number of walkers entering any site x equals the number leaving it. However, at a boundary site $y \in \partial K$, walkers arrive from neighbors but do not continue; they form the cluster. The operator $\mathcal{L}u(y)$ represents this absorption rate:

$$\text{Rate into } y \propto \sum_{z \sim y} (\text{Density at } z) = \sum_{z \sim y} u(z) = \Phi(y). \tag{2.31}$$

Since the harmonic measure $H(y)$ is the proportion of all walkers that terminate at y, it is directly proportional to this local absorption rate, i.e., the flux $\Phi(y)$.

This concludes the discrete formulation of the DLA process. The growth mechanism can be summarized as follows:

1. The system is governed by a harmonic potential $u(x)$ determined by the geometry of the cluster K.

2. This potential is the solution to the discrete Laplace equation $\mathcal{L}u = 0$ with $u|_K = 0$ and $u(x) \sim \ln |x|$ as $|x| \to \infty$.

3. The growth probability at a boundary site y is determined by the local flux $\Phi(y) = \sum_{z \sim y} u(z)$, where a new particle is added with probability $H(y) \propto \Phi(y)$.

2.3 Continuous Potential Theory

Much of this section parallels the discrete theory presented in section 2.2, though the derivations here require the delicate machinery of stochastic calculus. However, for our purposes, an understanding of the main results is sufficient to build a robust intuition for DLA. The fundamental shift is replacing the discrete random walker with its scaling limit: Brownian motion.

2.3.1 Brownian Motion and the Laplacian

We start by defining Brownian motion in both $\mathbb{R}^d$ and $\mathbb{C}$.

Definition 2.15 (Standard Brownian Motion)**.** *A standard Brownian motion in $\mathbb{R}^d$ is a stochastic process $B = (B_t)_{t\geq 0}$ defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ taking values in $\mathbb{R}^d$, such that:*

1. *$B_0 = 0$ almost surely.*
2. *The process has independent increments: for any $0 \leq s < t$, $B_t - B_s$ is independent of $\mathcal{F}_s = \sigma(B_u : u \leq s)$.*
3. *The increments are Gaussian: $B_t - B_s \sim \mathcal{N}(0, (t-s)I_d)$.*
4. *The paths $t \mapsto B_t$ are continuous almost surely.*

In the study of 2D DLA, we will utilise results from complex analysis. Thus, it is natural to identify the plane $\mathbb{R}^2$ with $\mathbb{C}$.

Definition 2.16 (Complex Brownian Motion)**.** *A canonical complex Brownian motion Z_t is a process taking values in $\mathbb{C}$, defined as:*

$$Z_t = X_t + iY_t, \tag{2.32}$$

where X_t and Y_t are two independent, standard one-dimensional Brownian motions.

The utility of this complex formulation becomes clear when we consider what is known as the quadratic variation of the process. In stochastic calculus, standard Brownian motions satisfy the formal rules $(dX_t)^2 = dt$ and $(dY_t)^2 = dt$. Applying this yields:

$$(dZ_t)^2 = (dX_t + idY_t)^2 = (dX_t)^2 - (dY_t)^2 + 2idX_tdY_t. \tag{2.33}$$

Since X and Y are independent identical copies, their variances match ($dt - dt = 0$), and the cross-term is negligible for variation. Thus, we effectively have $(dZ_t)^2 = 0$. This property identifies Z_t as a conformal martingale, which is the stochastic equivalent of an analytic function (where derivative depends only on z, not $\bar{z}$, as given by Wirtinger derivatives), and will be why we get the conformal invariance result later.

Remark 2.3. *Much of this section has been somewhat hand-wavy. However, for our purposes a complete rigourous introduction to stochastic calculus is unnecessary. If nothing else, think of Brownian motion as the continuous limit of a random walk. In fact, this has been rigorously proven as Donsker's Invariance Principle. For DLA, this implies that large clusters formed by random walks should exhibit the same macroscopic properties as those formed by Brownian particles.*

The connection between the physical process (Brownian motion) and the geometric potential (Laplace's equation) is provided by the infinitesimal generator.

Definition 2.17 (Continuous Generator and Laplacian)**.** *The Laplace operator (or Laplacian) Δ in $\mathbb{R}^d$ is the sum of the second partial derivatives:*

$$\Delta = \sum_{i=1}^{d} \frac{\partial^2}{\partial x_i^2}. \tag{2.34}$$

The infinitesimal generator of standard Brownian motion is the operator $\mathcal{A} = \frac{1}{2}\Delta$. It characterises the process via the martingale property: for any compactly supported C^2 (twice continuously differentiable) function f, the quantity

$$M_t = f(B_t) - f(B_0) - \int_0^t \frac{1}{2}\Delta f(B_s)\, ds \tag{2.35}$$

is a martingale with respect to the filtration of B_t.

This definition can be understood as a stochastic analogue to the Fundamental Theorem of Calculus. The integral term represents the expected drift of the function along the path. By subtracting this drift, we isolate the centered random noise, which is a martingale.

Remark 2.4 (Scaling Factors and Independence)**.** *We point out here that the scaling factor here differs from that in section 2.2.1. This discrepancy stems from the second-order term of the Taylor expansion used to derive the Brownian limit. However, since the potential distribution for DLA is determined by the condition $\mathcal{L}u = 0$ (or $\Delta u = 0$), this constant factor cancels out. The resulting harmonic measure is invariant. For a full derivation of this limit, see Lawler and Limic [5, Section 2.1].*

Definition 2.18 (Harmonic Functions)**.** *A twice continuously differentiable function $u : D \to \mathbb{R}$ defined on a domain $D \subset \mathbb{R}^d$ is called harmonic if it lies in the kernel of the Laplace operator:*

$$\Delta u(x) = 0 \quad \text{for all } x \in D. \tag{2.36}$$

By comparing this with the generator definition above, we see immediately that if u is harmonic, the drift term vanishes. This yields the critical result that $u(B_t)$ is a martingale, implying that the "average value" of a harmonic function is conserved along Brownian paths. We formalise this as the mean value property.

Theorem 2.6 (Mean Value Property)**.** *A locally integrable function $u : D \to \mathbb{R}$ is harmonic ($\Delta u = 0$) if and only if for every ball $B(x, r)$ strictly contained in the domain D:*

$$u(x) = \frac{1}{|\partial B(x,r)|} \int_{\partial B(x,r)} u(y)\, d\sigma(y), \tag{2.37}$$

where σ denotes the surface measure on the boundary sphere.

This is the continuous equivalent to the 'average of its neighbours' property of in the discrete setting.

2.3.2 Harmonic Measure and the Poisson Kernel

Before we present the continuous analogue to theorem 2.4, we define the continuous version of the harmonic measure.

Definition 2.19 (Harmonic Measure)**.** *Let $D \subset \mathbb{R}^d$ be a domain. The harmonic measure $\omega_x = \omega_x(D, \cdot)$ is the probability measure on the boundary ∂D defined by:*

$$\omega_x(E) = \mathbb{P}^x(B_{\tau_D} \in E), \tag{2.38}$$

for Borel sets $E \subset \partial D$, where $\tau_D = \inf\{t > 0 : B_t \notin D\}$ is the first exit time.

The solution to the continuous Dirichlet problem is given by Kakutani's Theorem.

Theorem 2.7 (Kakutani's Theorem)**.** *If D is a bounded domain and g is a continuous function on ∂D, then the function defined by:*

$$u(x) = \mathbb{E}^x[g(B_{\tau_D})] = \int_{\partial D} g(y)\, d\omega_x(y) \tag{2.39}$$

is harmonic in D and satisfies $\lim_{x \to y} u(x) = g(y)$ for regular boundary points $y \in \partial D$. Here, ω_x denotes the harmonic measure relative to x, representing the distribution of the first hitting point of Brownian motion on ∂D.

Proof. For a rigorous proof relying on the martingale property of harmonic functions, see Mörters and Peres [14, Theorem 3.12]. □

As in the discrete case, we are concerned with extending this to infinity. As a consequence of the following lemma, 2D Brownian motion is neighbourhood recurrent, meaning it will revisit any neighborhood infinitely often. This prevents us from simply imposing a boundary condition at infinity, just as recurrence did in the discrete case.

Lemma 2.1 (Exit Probabilities from an Annulus)**.** *Let B_t be a complex Brownian motion starting at z_0 within the annulus $A = \{z \in \mathbb{C} : r < |z| < R\}$. Let $T_A = \inf\{t > 0 : B_t \notin A\}$ be the first exit time. The probability that the process exits through the inner boundary $|z| = r$ is given by:*

$$\mathbb{P}^{z_0}(|B_{T_A}| = r) = \frac{\ln R - \ln |z_0|}{\ln R - \ln r}. \tag{2.40}$$

Proof. See Cass [15, Lemma 43]. The proof uses Itô's formula (stochastic chain rule) on the function $\ln |Z_t|$. □

Corollary 2.1 (Recurrence)**.** *Planar Brownian motion is neighborhood recurrent. For any starting point z_0 and any $\epsilon > 0$, the process will enter the disk $\{z : |z - z_0| < \epsilon\}$ almost surely:*

$$\mathbb{P}(\exists t > 0 : |B_t - z_0| < \epsilon) = 1. \tag{2.41}$$

Proof. Taking the limit $R \to \infty$ in the annulus exit probability yields $\mathbb{P}(|B_{T_A}| = r) \to 1$. □

2.3.3 Conformal Invariance

While in the discrete case we resolved this recurrence issue by turning to (regularised) Green's functions, we will instead utilise a fundamental property of Brownian motion: conformal invariance.

Theorem 2.8 (Lévy's Conformal Invariance Theorem)**.** *Let $U \subset \mathbb{C}$ be a domain and $f : U \to V$ be a non-constant analytic function. If Z_t is a complex Brownian motion in U, then the image process $W_t = f(Z_t)$ is, up to a random time change, a complex Brownian motion in V. Specifically, if $\tau_t = \int_0^t |f'(Z_s)|^2 ds$, then $f(Z_{\tau^{-1}(t)})$ is a standard Brownian motion in V.*

Proof. See Cass [15, Theorem 40]. □

Corollary 2.2 (Conformal Invariance of Harmonic Measure)**.** *Let $f : D \to D'$ be a conformal map. The harmonic measure is invariant under f:*

$$\omega_z(D, E) = \omega_{f(z)}(D', f(E)) \tag{2.42}$$

for any Borel set $E \subset \partial D$.

This result guarantees that utilising conformal maps in the sampling process will not distort the underlying harmonic measure, preserving the integrity of our off-lattice algorithm in the subsequent section.

2.3.4 Solvable Geometries

We conclude by introducing the probability density function for the hitting probabilities, and finding it explicitly for two fundamental domains used in DLA algorithms: the Upper Half-Plane ($\mathbb{H}$) and the Unit Disk ($\mathbb{D}$).

Definition 2.20 (Poisson Kernel)**.** *Let σ be the surface measure on ∂D. If the harmonic measure is absolutely continuous with respect to ζ, the Poisson kernel $P(x, y)$ is defined as the Radon-Nikodym derivative:*

$$P(x, y) = \frac{d\omega_x}{d\sigma}(y). \tag{2.43}$$

It represents the probability density of the Brownian motion exiting D at y given it started at x.

We can now express the conformal invariance of the harmonic measure in terms of the Poisson kernel. By definition, the harmonic measure of a set $E \subset \partial D$ is the integral of the Poisson kernel with respect to the surface measure $d\sigma$:

$$\omega_z(D, E) = \int_E P_D(z, \zeta) d\sigma(\zeta). \tag{2.44}$$

Applying the conformal invariance corollary to a map $f : D \to \mathbb{D}$ such that $f(z) = 0$, and noting that the Poisson kernel on the disk from the origin is uniform ($P_{\mathbb{D}}(0, w) = \frac{1}{2\pi}$), we have:

$$\int_E P_D(z, \zeta) d\sigma(\zeta) = \omega_0(\mathbb{D}, f(E)) = \int_{f(E)} \frac{1}{2\pi} d\sigma(w). \tag{2.45}$$

By the change of variables $w = f(\zeta)$, the arc length element transforms as $d\sigma(w) = |f'(\zeta)|d\sigma(\zeta)$, resulting in:

$$\int_E P_D(z,\zeta)d\sigma(\zeta) = \int_E \frac{1}{2\pi}|f'(\zeta)|d\sigma(\zeta). \tag{2.46}$$

This equality holds for any Borel set E, hence:

$$P_D(z,\zeta) = \frac{1}{2\pi}|f'(\zeta)|. \tag{2.47}$$

We now apply this relation to our domains of interest:

Theorem 2.9 (Poisson Kernel for $\mathbb{H}$)**.** *Consider the upper half-plane $\mathbb{H} = \{z \in \mathbb{C} : Im(z) > 0\}$. The probability density of a Brownian particle starting at $z = x + iy$ hitting the real line at $u \in \mathbb{R}$ is given by the Cauchy distribution:*

$$P_{\mathbb{H}}(z,u) = \frac{1}{\pi}\frac{y}{(u-x)^2 + y^2}. \tag{2.48}$$

Proof. We construct a conformal map $\phi : \mathbb{H} \to \mathbb{D}$ that sends the starting point z to the origin 0, and the real line $\mathbb{R}$ to the unit disk. The standard choice is the Cayley transform variation:

$$\phi(w) = \frac{w-z}{w-\bar{z}}. \tag{2.49}$$

Observe that $\phi(z) = 0$, and for any real boundary point $u \in \mathbb{R}$, $|\phi(u)| = 1$. Using the pullback proposition, the kernel is given by $\frac{1}{2\pi}|\phi'(w)|$. We compute the derivative with respect to the w:

$$\phi'(w) = \frac{z-\bar{z}}{(w-\bar{z})^2}. \tag{2.50}$$

Now evaluated at a boundary point $w = u$:

$$|\phi'(u)| = \frac{|2iy|}{|u-\bar{z}|^2} = \frac{2y}{(u-x)^2 + y^2}. \tag{2.51}$$

Thus, the Poisson kernel is:

$$P_{\mathbb{H}}(z,u) = \frac{1}{2\pi}\left(\frac{2y}{(u-x)^2 + y^2}\right) = \frac{1}{\pi}\frac{y}{(u-x)^2 + y^2}. \tag{2.52}$$

□

We now follow the same procedure for the unit disk.

Theorem 2.10 (Poisson Kernel for $\mathbb{D}$)**.** *The Poisson kernel for the unit disk $\mathbb{D}$, representing the exit density at angle θ for a particle starting at $z \in \mathbb{D}$, is given by:*

$$P_{\mathbb{D}}(z,\theta) = \frac{1}{2\pi}\frac{1-|z|^2}{|e^{i\theta} - z|^2}. \tag{2.53}$$

Proof. We seek an automorphism of the disk $\psi : \mathbb{D} \to \mathbb{D}$ that maps the starting point z to the

origin 0. This is given by the Möbius transformation:

$$\psi(w) = \frac{w - z}{1 - \bar{z}w}. \tag{2.54}$$

The derivative at the boundary point w is:

$$\psi'(w) = \frac{1 - |z|^2}{(1 - \bar{z}w)^2}. \tag{2.55}$$

To find the magnitude $|\psi'(w)|$, we observe that for $w = e^{i\theta}$, the denominator satisfies:

$$|1 - \bar{z}e^{i\theta}| = |e^{-i\theta} - \bar{z}| = |\overline{e^{-i\theta} - \bar{z}}| = |e^{i\theta} - z|. \tag{2.56}$$

Therefore, the Poisson kernel is:

$$\frac{1}{2\pi}|\psi'(e^{i\theta})| = \frac{1}{2\pi}\frac{1 - |z|^2}{|e^{i\theta} - z|^2}. \tag{2.57}$$

□

This concludes the continuous formulation of the DLA process. The theoretical framework parallels the discrete setting but utilises slightly different machinery:

1. The system is governed by a continuous potential field $u(x)$ determined by the geometry of the cluster, representing the probability density of the underlying Brownian motion.

2. $u(x)$ satisfies the continuous Laplace equation $\Delta u = 0$, but solving this explicitly is difficult. Instead we utilise the property of conformal invariance, allowing us to encode the potential field geometrically.

3. The growth probability at a boundary point is given by the Poisson Kernel, which serves as the continuous analogue to the discrete flux.

Chapter 3

Algorithms

3.1 Introduction

We discuss why naive simulation of DLA is inefficient and introduce the three algorithmic approaches implemented in this study: On-Lattice, Off-Lattice, and the Hybrid model.

3.2 On-Lattice Implementation

3.2.1 The Fuzzy Annulus Method

3.2.2 Walk-to-Line

3.2.3 Walk-to-Square

3.3 Off-Lattice Implementation

3.3.1 Optimization Strategies

Remark 3.1 (Justification for DLA Initialization)**.** *If we start the Brownian motion at the center of the disk ($z = 0$), the kernel simplifies to:*

$$P_{\mathbb{D}}(0, \theta) = \frac{1}{2\pi}. \tag{3.1}$$

This constant density proves that a random walker starting at the origin hits the bounding circle ***uniformly****. This rigorously justifies the standard DLA algorithm initialization, where particles are spawned at uniformly random angles on a large bounding circle.*

3.3.2 Exact Collision

3.3.3 Comparison to On-Lattice

3.4 The Hybrid Model

3.4.1 Motivation

3.4.2 Theoretical Alternatives

3.4.3 Selection and Justification

3.4.4 Implementation

We define the specific "snapping" mechanism: particles propagate in continuous $\mathbb{R}^2$ but aggregate to $\mathbb{Z}^2$.

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Appendix A

Algorithmic Details and Optimizations

A.1 Lattice Core Optimizations

A.2 Off-Lattice and Hybrid Details

A.3 Data Structures

Appendix B

Reference: LaTeX Structure Examples

This section provides templates for common mathematical structures used in this dissertation.

B.1 Aligned Equations

Below is a multi-line derivation showing the alignment of the equals sign using the `align` environment.

$$\begin{aligned} \nabla^2\phi(r,\theta) &= \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial\phi}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2\phi}{\partial\theta^2} && \text{(B.1)} \\ &= 0 \quad \text{(Laplace Equation)} \\ \implies \phi(r,\theta) &= \sum_{n=0}^{\infty} r^n (A_n \cos n\theta + B_n \sin n\theta). && \text{(B.2)} \end{aligned}$$

B.2 Floats

B.2.1 Single Figure

Figure B.1 shows a standard inclusion of a graphic.

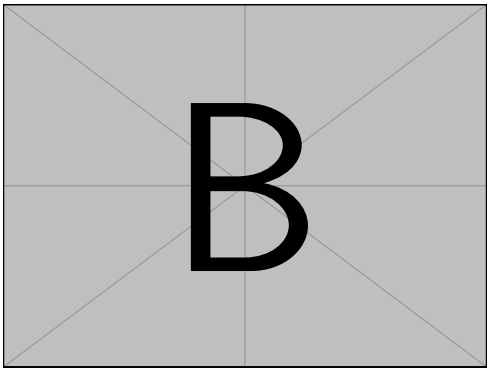
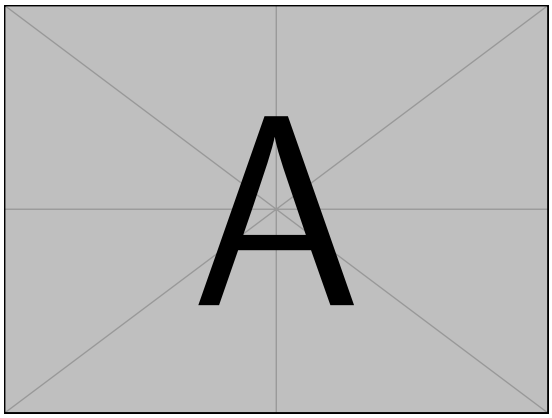


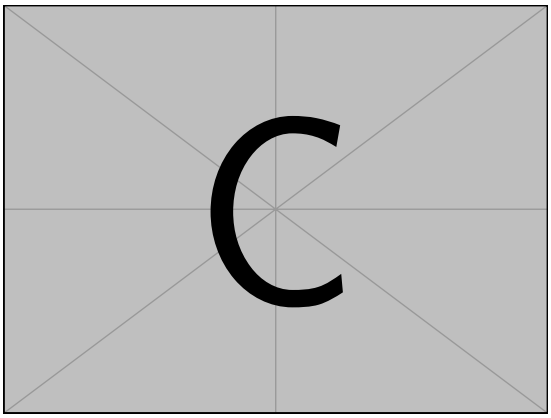
Figure B.1: A realization of 2D Brownian motion.

B.2.2 Side-by-Side Figures (Subfloats)

Figure B.2 demonstrates how to place two plots next to each other for comparison.



(a) On-Lattice DLA



(b) Off-Lattice DLA

Figure B.2: Comparison of cluster morphology between lattice and continuum models.

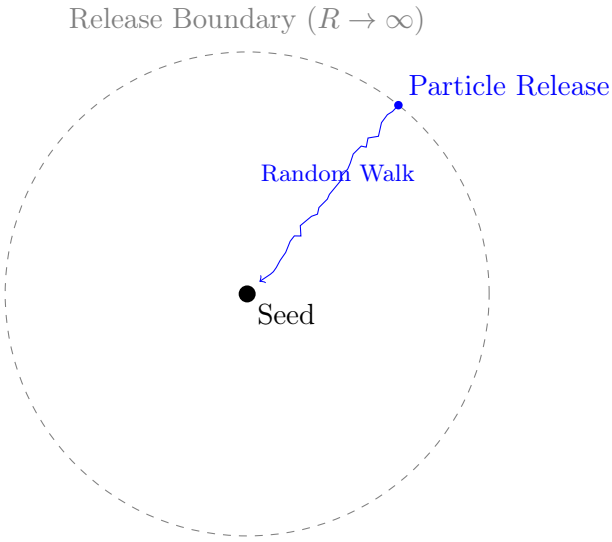


Figure B.3: Schematic of the DLA growth process.

B.3 Algorithms

Algorithm 1 shows a basic random walk loop using the `algorithm2e` package.

Algorithm 1: Basic Random Walk Simulation

Result: Final position of particle

```
Initialize particle at r_start;
while particle not stuck AND particle within bounds do
    Generate random angle θ ∈ [0, 2π);
    x ← x + δ cos θ;
    y ← y + δ sin θ;
    if distance to cluster < ϵ then
        Stick particle;
        break;
    end
end
```

B.4 Theorems

Standard environments for formal mathematical statements.

Definition B.1 (Harmonic Measure)**.** *The harmonic measure $\omega(z, E, \Omega)$ is the probability that a Brownian motion starting at $z \in \Omega$ first hits the boundary $\partial\Omega$ in the set $E \subset \partial\Omega$.*

Theorem B.1 (Kakutani's Theorem)**.** *If D is a bounded domain and g is a continuous function on ∂D, then the function defined by:*

$$u(x) = \mathbb{E}^x[g(B_{\tau_D})] = \int_{\partial D} g(y)\, d\omega_x(y) \tag{B.3}$$

is harmonic in D and satisfies $\lim_{x \to y} u(x) = g(y)$ for regular boundary points $y \in \partial D$.

B.5 References

This is an example of a citation. Witten and Sander introduced the model in 1981 [1].